

EC-Council



Digital Forensics Essentials



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MODULE 6

Network Forensics



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Network Forensics Fundamentals

SUBTOPIC 1



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Module Objectives

- 1 Understanding the Network Forensics Fundamentals
- 2 Understanding the Logging Fundamentals
- 3 Understanding the Event Correlation Concepts
- 4 Overview of Identifying Indicators of Compromise (IoCs) from Network Logs
- 5 Understanding How to Investigate Network Traffic



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Module Flow



1

**Understand Network
Forensics
Fundamentals**



2

**Understand Event
Correlation Concepts
and Types**



3

**Identify Indicators of
Compromise (IoCs)
from Network Logs**



4

**Investigate Network
Traffic**



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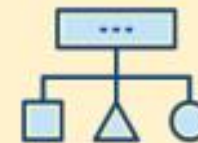


Introduction to Network Forensics

- ❑ Network forensics is the **capturing, recording, and analysis** of **network events** in order to discover the source of security incidents

Network forensics can reveal the following information:

- ✓ Source of security incidents
- ✓ The path of intrusion
- ✓ The Intrusion techniques an attacker used
- ✓ Traces and evidence



Postmortem and Real-Time Analysis

Forensic examination of logs can be divided into two categories :



Postmortem

- ❑ Postmortem analysis of logs is conducted to investigate an incident that **has already happened**



Real-Time Analysis

- ❑ Real-time analysis is conducted to detect and examine an **ongoing-attack**



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Network Attacks

Most common attacks on networks

- Eavesdropping
- Data Modification
- IP Address Spoofing
- Denial-of-Service Attack
- Man-in-the-Middle Attack
- Packet Sniffing
- Enumeration
- Session Hijacking
- Buffer Overflow
- Email Infection
- Malware attack
- Password-based attack
- Router Attacks

Attacks specific to wireless networks

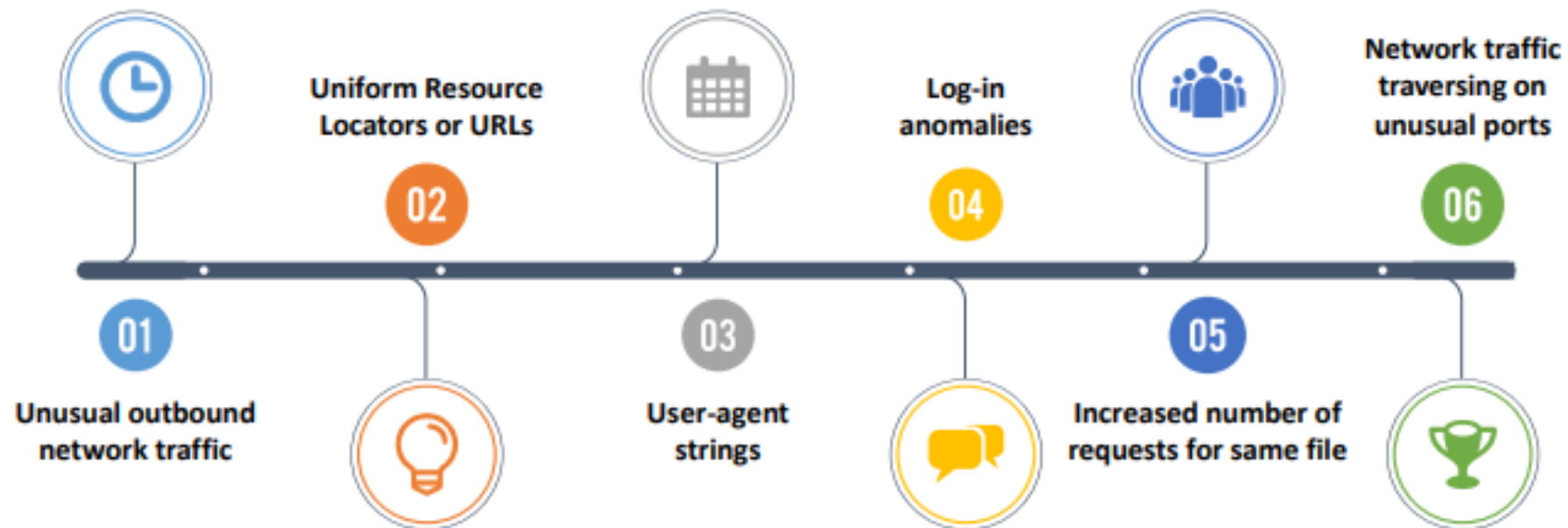
- Rogue Access Point Attack
- Client Misassociation
- Misconfigured Access Point Attack
- Unauthorized Association
- Ad Hoc Connection Attack
- Honeypot Access Point Attack
- AP MAC Spoofing
- Jamming Attack



Indicators of Compromise (IOCs)

- ❑ Indicators of Compromise (IoCs) are **digital** forensic **artifacts** that help detect a security incident that **has occurred** (or **is ongoing**) on a host system or a network
- ❑ These artifacts include logs related to systems, applications, networks, firewalls, etc.

Examples of IoCs:



Where to Look for Evidence



Logs collected in **network devices** and **applications** can be used as evidence for investigating network security incidents

Functions

Handles high-level protocols, issues of representation, encoding, and dialog control

Provides a logical connection between the endpoints and provides transport

Selects the best path through the network for data flow

Defines how to transmit an IP datagram to the other devices

Layers of TCP/IP Model

Application layer



Transport layer



Internet layer



Network Access layer

Protocols

File Transfer (TFTP, FTP, NFS), Email (SMTP), Network Management (SNMP), Name Management (DNS)

Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)

Internet Protocol (IP), Internet Control Message Protocol (ICMP), Address Resolution Protocol (ARP)

Ethernet, Fast Ethernet, SLIP, PPP, FDDI, ATM, Frame Relay, SMDS, ARP, Proxy ARP, RARP

Network Devices and Applications Logs

Servers/Desktops, Anti-virus, Business Applications, Databases

Firewall, IDS/IPS

Firewall, IDS/IPS, VPN

Routers and Switches



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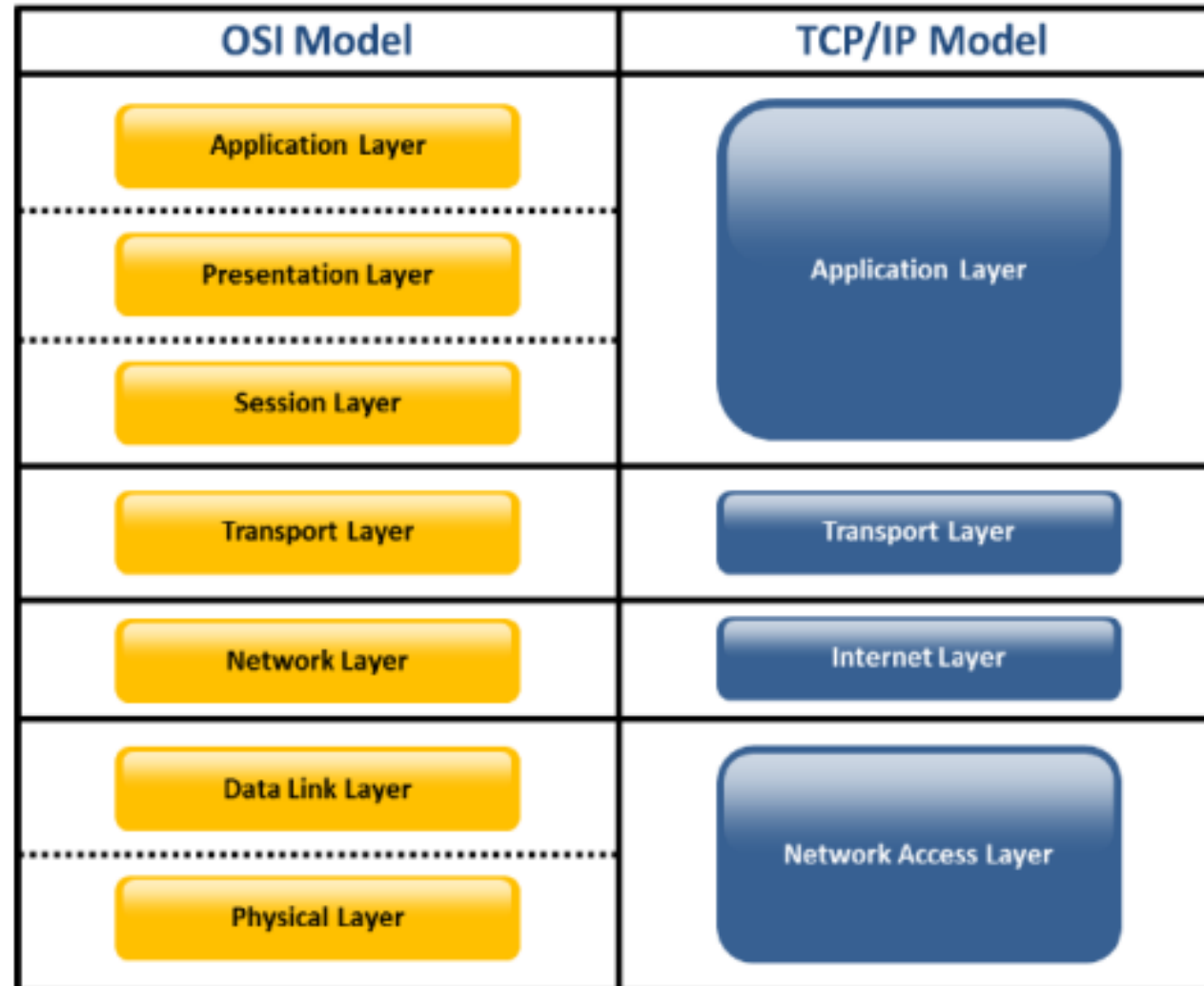


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OSI Model vs. TCP/IP Model

Types of Network-based Evidence

Full Content Data

- ❖ Full content data refers to **actual packets** that are collected by storing the network traffic (known as packet capture or PCAP files)
- ❖ Investigators can use tools like **tcpdump** or **Wireshark** to analyze any subset of full content data

Session Data

- ❖ Session data refers to a summary of **conversation** between **two network entities**
- ❖ It includes details such as destination IP and destination port, source IP and source port, start time of the session, and amount of information exchanged during the session

Alert Data

- ❖ Alert data is triggered by tools such as **Snort IDS** and **Suricata** that are preprogrammed to examine network traffic for IoCs and report the findings as alerts
- ❖ Investigators need to be careful while examining alert data as there might be **false positive** alerts

Statistical Data

- ❖ Statistical data provides overall profile or **summaries** of the **network traffic**
- ❖ Statistical data analysis can provide investigator with useful information such as **timestamps** related to network conversations, **protocols** and **services** being used, average **packet size**, and average **packet rate**



Module Flow



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3

**Identify Indicators of
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Event Correlation

1

Event correlation is the process of relating a set of events that have occurred in a **predefined interval of time**

2

It includes analysis of events to determine how it could add up to become a bigger event

3

It is usually performed on a **log management platform**, after identifying logs having similar properties

Steps in event correlation:

Event aggregation

Event masking

Event filtering

Root cause analysis



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Types of Event Correlation

Same-Platform Correlation

- ❑ This correlation method is used when **one common OS** is used throughout the network in an organization
- ❑ **Example:** An organization running only Microsoft Windows OS (any version) on their servers may **collect event log entries** and perform **trend analysis diagonally**



Cross-Platform Correlation

- ❑ This correlation method is used when **different OS and network hardware platforms** are used in the network of an organization
- ❑ **Example:** Clients may use Microsoft Windows, but they use a Linux-based firewall and email gateway

Prerequisites of Event Correlation

Transmission of Data

- ☐ Transmitting log data from one security device to another until it **reaches a consolidation point in the automated system**
- ☐ To have a secure transmission and reduce the risk of exposure during transmission, the data must be **encrypted and authenticated**

Normalization

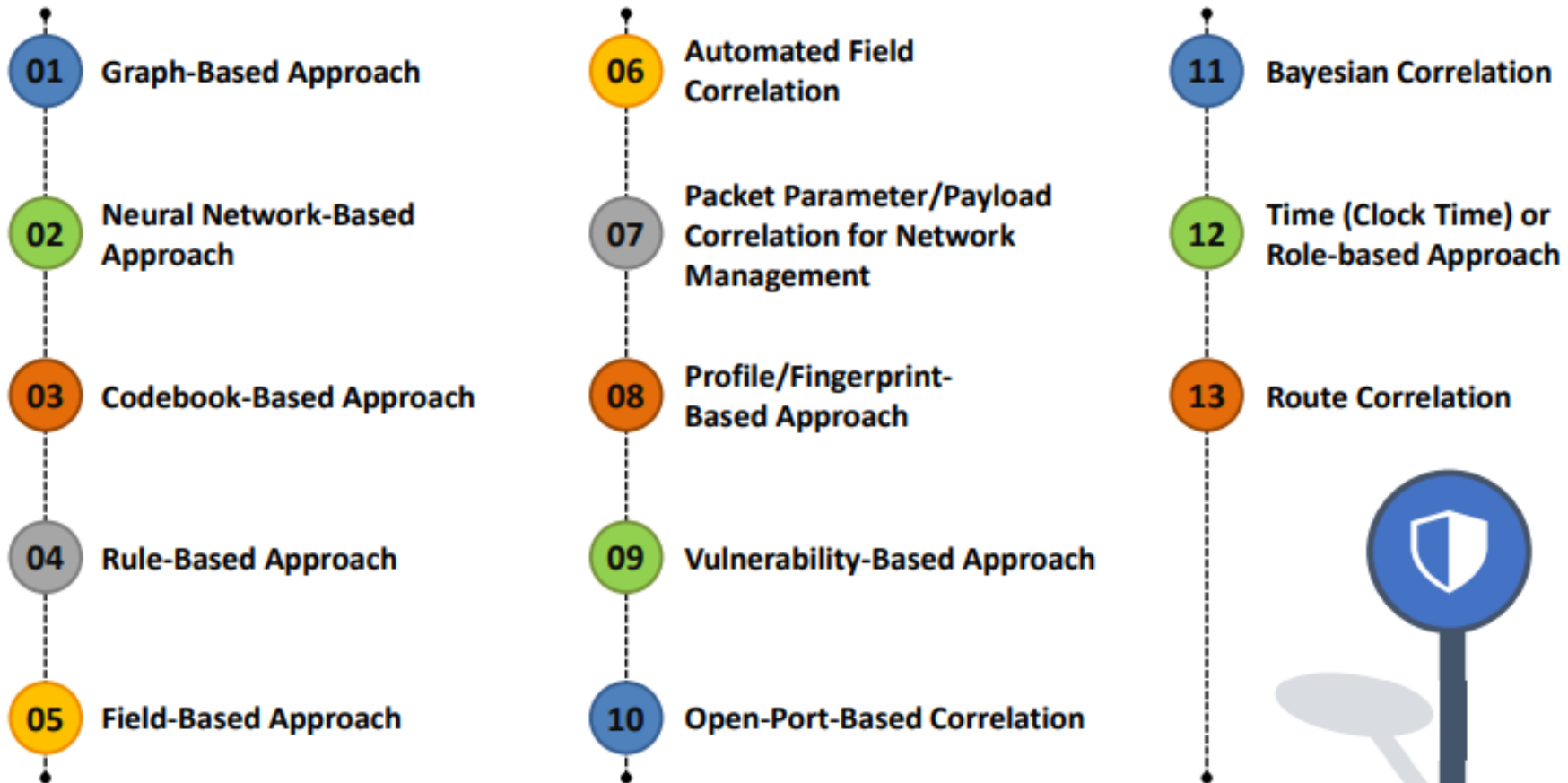
- ☐ After the data is gathered, it must be **formatted** again from different log formats to a single or polymorphic log that can be easily inserted into the database



Data Reduction

- ☐ After collecting the data, repeated data must be **removed** so that the data can be correlated more efficiently
- ☐ Removing unnecessary data can be done by **compressing the data, deleting repeated data, filtering**, or combining similar events into a single event, and sending that to the correlation engine

Event Correlation Approaches



Identify Indicators of Compromise (IoCs) from Network Logs

SUBTOPIC 2



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Analyzing Firewall Logs



Firewalls are the **first points of entry into a network** and store details of all the data packets moving in and out of the network



The network **firewall logs collect network traffic data** such as request source and destination, ports used, time and date, and priority



These details help investigators **correlate the data with other suspicious files** to identify the source and other targets of an attack



Investigators need to analyze the logs carefully based on the **timings and suspicious IP addresses**



Check for **application generated requests, DNS queries, suspicious IP addresses, and URLs**



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Analyzing Firewall Logs: Cisco

- ❑ Cisco Firewall uses **mnemonics** as identifiers to represent severity of an event

Mnemonic	Severity	Description
4000nn	4	IPS:number string from IP_address to IP_address on interface interface_name
106001	2	Inbound TCP connection denied from IP_address/port to IP_address/port flags tcp_flags on interface interface_name
106002	2	protocol Connection denied by outbound list acl_ID src inside_address dest outside_address
106006	2	Deny inbound UDP from outside_address/outside_port to inside_address/inside_port on interface interface_name
106007	2	Deny inbound UDP from outside_address/outside_port to inside_address/inside_port due to DNS (Response Query)
106010	3	Deny inbound protocol src interface_name:dest_address/dest_port dst
106012	3	Deny IP from IP_address to IP_address, IP options hex
106013	3	Dropping echo request from IP_address to PAT address IP_address
106014	3	Deny inbound icmp src interface_name:IP_address dst interface_name: IP_address (type dec, code dec)
106015	6	Deny TCP (no connection) from IP_address/port to IP_address/port flags tcp_flags on interface interface_name
106016	2	Deny IP spoof from (IP_address) to IP_address on interface interface_name.
106017	2	Deny IP due to Land Attack from IP_address to IP_address
106018	2	ICMP packet type ICMP_type denied by outbound list acl_ID src inside_address dest outside_address
106020	2	Deny IP teardrop fragment (size = number, offset = number) from IP_address to IP_address
106021	1	Deny protocol reverse path check from source_address to dest_address on interface interface_name
106022	1	Deny protocol connection spoof from source_address to dest_address on interface interface_name
106023	4	Deny protocol src [interface_name:source_address/source_port] dst interface_name:dest_address/dest_port [type {string}, code {code}] by access_group acl_ID
106100	4	access-list acl_ID [permitted denied est-allowed] protocol interface_name/source_address(source_port) -> interface_name/dest_address(dest_port) hit-cnt number ({first hit number-second interval})
710003	3	{TCP UDP} access denied by ACL from source_IP/source_port to interface_name:dest_IP/service



Analyzing Firewall Logs: Check Point

- ❑ Check Point firewall logs can be viewed through a **Check Point Log viewer**
- ❑ It uses icons and colors in the log table to represent different security events and their severity
- ❑ **Red** represents the **connection attempts blocked** by the **firewall**
- ❑ **Orange** signifies **traffic detected** as suspicious, but accepted by the firewall
- ❑ **Green** is for the **traffic accepted** by the **firewall**

Some icons used in Check Point logs

Action	Icon	Description
Connection Accepted		The firewall accepted a connection
Connection Decrypted		The firewall decrypted a connection
Connection Dropped		The firewall dropped a connection
Connection Encrypted		The firewall encrypted a connection
Connection Rejected		The firewall rejected a connection
Connection Monitored		A security event was monitored; however, it was not blocked, due to the current configuration
URL Allowed		The firewall allowed a URL
URL Filtered		The firewall blocked a URL
Virus Detected		A virus was detected in an email
Potential Spam Stamped		An email was marked as potential spam
Potential Spam Detected		An email was rejected as potential spam
Mail Allowed		A non-spam email was logged
Blocked by VStream Antivirus		VStream Antivirus blocked a connection

Analyzing IDS Logs



- ❑ Intrusion Detection System (IDS) logs provide information helpful in finding **suspicious packet** types, determining probes, generating new attack signatures, and measuring attack statistics
- ❑ Some of the common **IDS devices** and **tools** include **Juniper**, **Check Point** and **Snort**

General indicators of intrusion:

1

Requests targeted towards known vulnerabilities

2

Failure to comply with protocols and syntaxes

3

Unexpected elements such as date, time, and system resources

Repeated unusual network activity **4**

Address anomalies in traffic **5**

Occurrence of mistyped command **6**



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Why Investigate Network Traffic?



- ☐ To detect and examine an **ongoing attack** by monitoring network traffic communication patterns
- ☐ To know which hosts or networks are **involved** in a network security incident
- ☐ To trace information/packets related to a **security intrusion** and collect them as evidence



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Gathering Evidence via Sniffers

Sniffer is a computer software or hardware that can **intercept** and **log traffic** passing over a network

Sniffers put NICs in **promiscuous mode** to allow them to listen to and capture all data transmitted over the network

Sniffers **collect traffic** from the **network** and **transport layers** other than the physical and data-link layer

Investigators should **configure sniffers** for the size of frames to be captured



Sniffing Tool: Tcpcdump

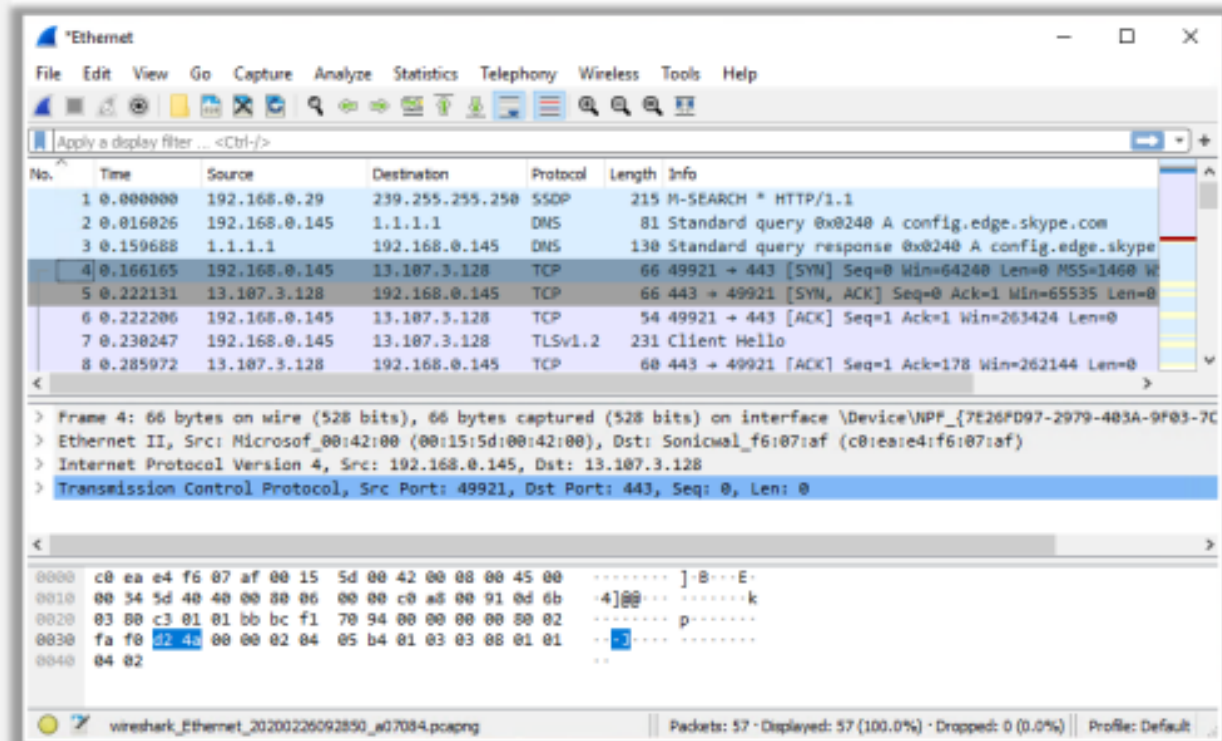
- ❑ Tcpdump is a **command line interface packet sniffer** which runs on Unix-based OSes

[illegible]

<https://www.fcpsdump.org>

Sniffing Tool: Wireshark

- ❑ It enables you to **capture** and **interactively browse the traffic** running on a computer network
- ❑ Wireshark uses **Winpcap** to capture packets, and therefore, can only capture packets on networks supported by Winpcap
- ❑ It **captures live network traffic** from Ethernet, IEEE 802.11, PPP/HDL, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks
- ❑ Captured files can be programmatically edited via **command-line**
- ❑ A **set of filters** for customized data display can be refined using a display filter



<https://www.wireshark.org>



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Analyze Traffic for SYN-FIN Flood DoS Attack

- ❑ Attackers send packets with both the SYN and FIN flags in an attempt to saturate the network bandwidth and cause a DoS attack
- ❑ Use the filter **tcp.flags==0x003** to detect a SYN/FIN flooding attack on Wireshark
- ❑ The screenshot below shows a large number of TCP SYN-FIN packets being passed from a single source IP address **10.0.0.2** to the destination IP address **10.128.0.2** on HTTP **port 80**

Wireshark packet capture showing a SYN-FIN flood attack. The filter is **tcp.flags==0x003**. The packet list shows multiple packets from source **10.0.0.2** to destination **10.128.0.2** on port **80**. The packet details pane shows the selected packet as a Transmission Control Protocol, Src Port: 2512, Dst Port: 80, Seq: 0, Len: 0.

No.	Time	Source	Destination	Protocol	Length	Info
19	0.153154	10.0.0.2	10.128.0.2	TCP	54	2521 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
20	0.161621	10.0.0.2	10.128.0.2	TCP	54	2522 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
21	0.170057	10.0.0.2	10.128.0.2	TCP	54	2523 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
22	0.178463	10.0.0.2	10.128.0.2	TCP	54	2524 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
23	0.186853	10.0.0.2	10.128.0.2	TCP	54	2525 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
24	0.195318	10.0.0.2	10.128.0.2	TCP	54	2526 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
25	0.203734	10.0.0.2	10.128.0.2	TCP	54	2527 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
26	0.212143	10.0.0.2	10.128.0.2	TCP	54	2528 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
27	0.220546	10.0.0.2	10.128.0.2	TCP	54	2529 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
28	0.228911	10.0.0.2	10.128.0.2	TCP	54	2530 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
29	0.237455	10.0.0.2	10.128.0.2	TCP	54	2531 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
30	0.245881	10.0.0.2	10.128.0.2	TCP	54	2532 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
31	0.254519	10.0.0.2	10.128.0.2	TCP	54	2533 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
32	0.262936	10.0.0.2	10.128.0.2	TCP	54	2534 → 80 [FIN, SYN] Seq=0 Win=512 Len=0
33	0.271378	10.0.0.2	10.128.0.2	TCP	54	2535 → 80 [FIN, SYN] Seq=0 Win=512 Len=0

- ❑ Navigate to **Statistics > Capture File Properties** to analyze the summary of packets captured
- ❑ The screenshot below shows that the **packet capture time** window is 14 seconds long and the average number of **packets sent per second** is 118 at 51Kbps speed, indicating a SYN-FIN flooding attack

Wireshark - Capture File Properties - SYN-FIN.pcap

Details

Interfaces

Interface	Captured packets	Capture filter	Link type	Packet size limit
Unknown	Unknown	Unknown	Ethernet	32767 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	1657	1657 (100.0%)	---
Time span, s	13.979	13.979	---
Average pps	118.5	118.5	---
Average packet size, B	54	54	---
Bytes	89478	89478 (100.0%)	0
Average bytes/s	6400	6400	---
Average bits/s	51 k	51 k	---

Capture file comments

Refresh <https://kb.mazebolt.com> Save Comments Close Copy To Clipboard Help



SYN-FIN.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags==0x003

No.	Time	1 Source	2 Destination	3 Protocol	Length	Info	4
19	0.153154	10.0.0.2	10.128.0.2	TCP	54	2521 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
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28	0.229011	10.0.0.2	10.128.0.2	TCP	54	2530 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
29	0.237455	10.0.0.2	10.128.0.2	TCP	54	2531 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
30	0.245881	10.0.0.2	10.128.0.2	TCP	54	2532 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
31	0.254519	10.0.0.2	10.128.0.2	TCP	54	2533 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
32	0.262936	10.0.0.2	10.128.0.2	TCP	54	2534 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0
33	0.271378	10.0.0.2	10.128.0.2	TCP	54	2535 → 80 [FIN, SYN]	Seq=0 Win=512 Len=0

Large number of SYN-FIN packets detected on Wireshark



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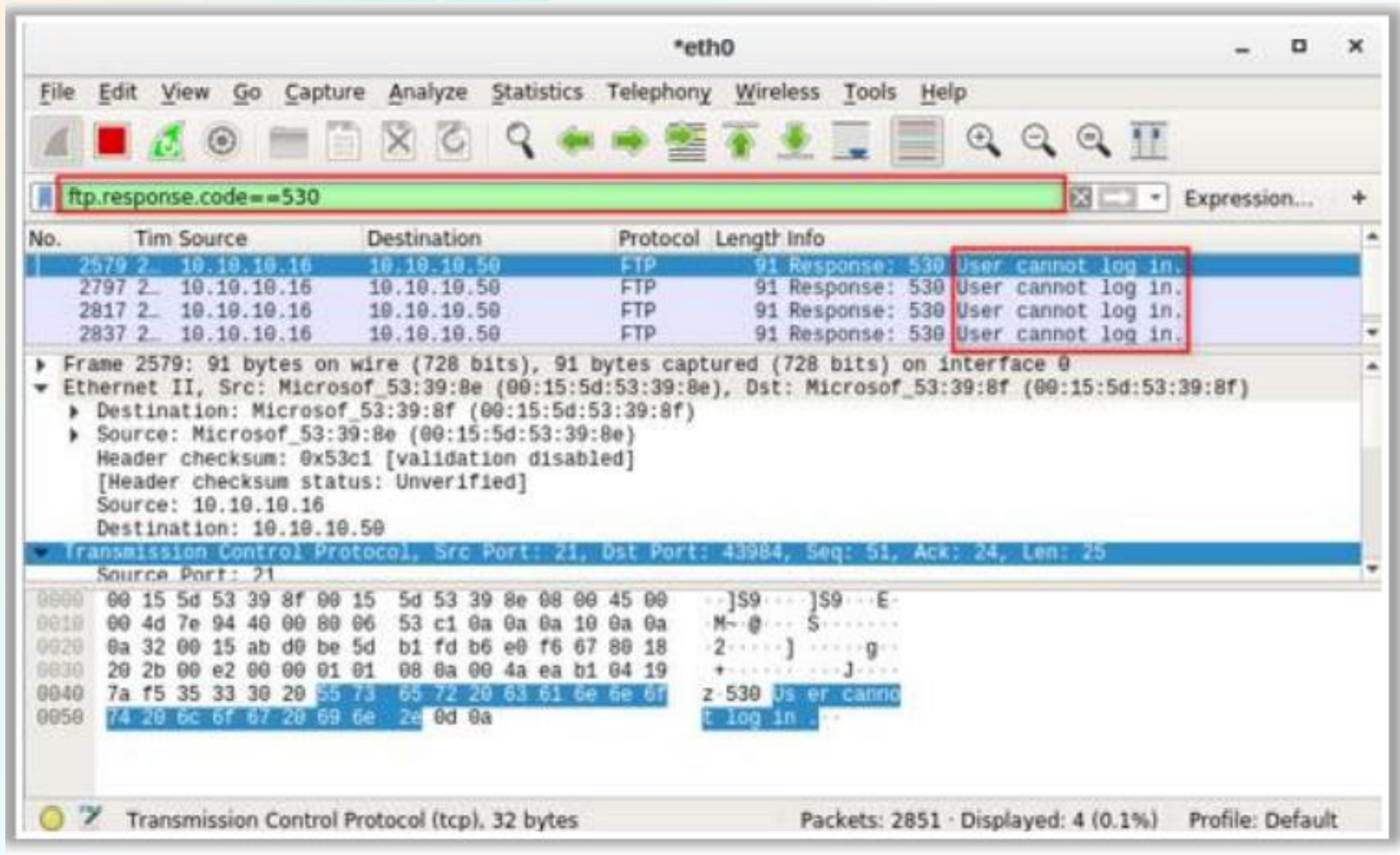
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Analyze Traffic for FTP Password Cracking Attempts

- ❑ In a password cracking attack, the attacker attempts to gain access to the credentials of an authenticated user via various techniques, such as brute-force or dictionary attacks
- ❑ Apply the filter `ftp.response.code == 530` to monitor all unsuccessful login attempts over FTP
- ❑ The screenshot above shows multiple **unsuccessful login attempts** from the **source IP 10.10.10.16** to the target IP 10.10.10.50, which is strongly indicative of a brute-force attack
- ❑ Apply the filter `ftp.response.code == 230` to see successful logins on the FTP server
- ❑ The screenshot below shows **two successful logins** from the source IP 10.10.10.16 which indicates that attacker has successfully obtained the credentials

The top screenshot shows a Wireshark capture on interface eth0 with the filter `ftp.response.code == 530`. It displays four packets (No. 2575, 2791, 2917, 2937) from source IP 10.10.10.16 to destination IP 10.10.10.50. All are FTP responses with code 530, indicating 'User cannot log in'. The bottom packet details show a 'User logged in' message.

The bottom screenshot shows the same capture with the filter changed to `ftp.response.code == 230`. It displays two packets (No. 2916, 2936) from source IP 10.10.10.16 to destination IP 10.10.10.50. Both are FTP responses with code 230, indicating 'User logged in'. The bottom packet details show a 'User logged in' message.

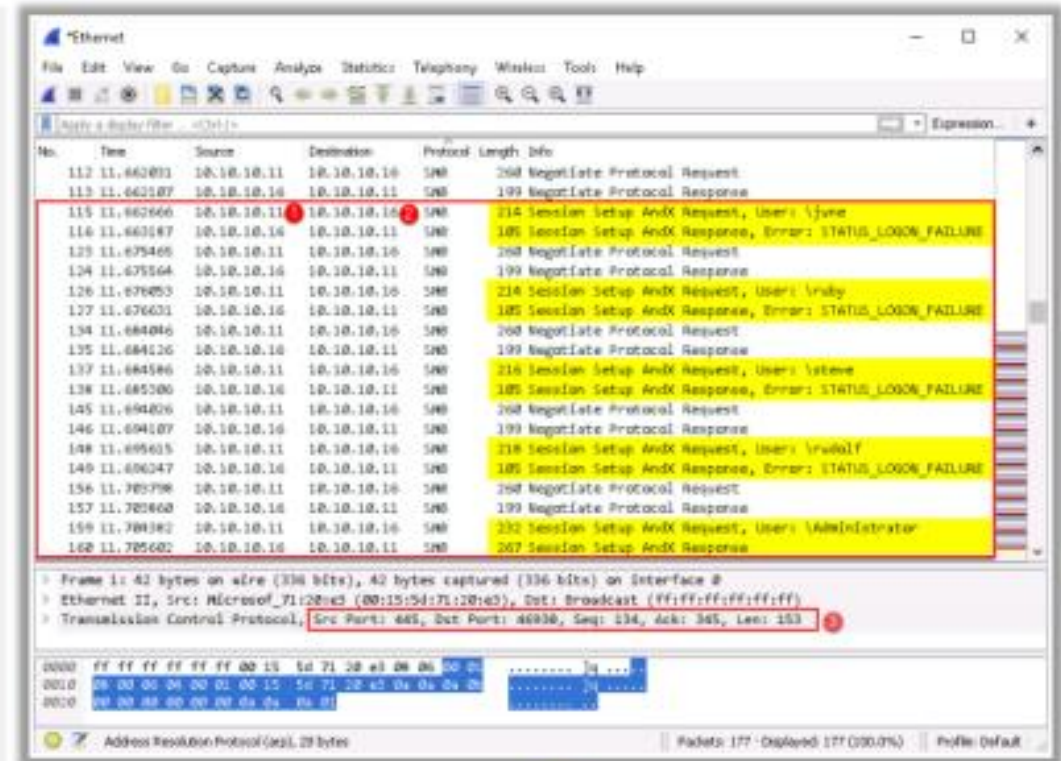


Multiple unsuccessful login attempts observed on FTP via Wireshark

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Analyze Traffic for SMB Password Cracking Attempts

- ❑ The screenshot shows that several login attempts on the **IP 10.10.10.16** of the target host (2) on SMB from the **Source IP 10.10.10.11** (1)
- ❑ The traffic captured via Wireshark reveals **several usernames** along with the message '**Error: STATUS_LOGON_FAILURE**' which strongly indicates a brute-force attack attempt on the SMB protocol
- ❑ If any name found on the captured packets matches that of an authorized user on the target host, you can consider that the **brute-force attack** is likely to have been successful
- ❑ Gather other useful **information** under the Transmission Control Protocol section such as **source port, destination port** and **count of packet bytes** (3)



No.	Time	Source	Destination	Protocol	Length	Info
112	11.662093	10.10.10.11	10.10.10.16	SMB	268	Negotiate Protocol Request
113	11.662107	10.10.10.16	10.10.10.11	SMB	199	Negotiate Protocol Response
115	11.662666	10.10.10.11	10.10.10.16	SMB	214	Session Setup AndX Request, User: /june
116	11.663167	10.10.10.16	10.10.10.11	SMB	185	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
125	11.675466	10.10.10.11	10.10.10.16	SMB	268	Negotiate Protocol Request
124	11.675564	10.10.10.16	10.10.10.11	SMB	199	Negotiate Protocol Response
126	11.676053	10.10.10.11	10.10.10.16	SMB	214	Session Setup AndX Request, User: /indy
127	11.676631	10.10.10.16	10.10.10.11	SMB	185	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
134	11.684046	10.10.10.11	10.10.10.16	SMB	268	Negotiate Protocol Request
135	11.684126	10.10.10.16	10.10.10.11	SMB	199	Negotiate Protocol Response
137	11.684586	10.10.10.11	10.10.10.16	SMB	214	Session Setup AndX Request, User: /steve
138	11.685300	10.10.10.16	10.10.10.11	SMB	185	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
145	11.694826	10.10.10.11	10.10.10.16	SMB	268	Negotiate Protocol Request
146	11.694107	10.10.10.16	10.10.10.11	SMB	199	Negotiate Protocol Response
148	11.695635	10.10.10.11	10.10.10.16	SMB	214	Session Setup AndX Request, User: /wulff
149	11.696347	10.10.10.16	10.10.10.11	SMB	185	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
156	11.703798	10.10.10.11	10.10.10.16	SMB	268	Negotiate Protocol Request
157	11.703868	10.10.10.16	10.10.10.11	SMB	199	Negotiate Protocol Response
159	11.709382	10.10.10.11	10.10.10.16	SMB	232	Session Setup AndX Request, User: /Administrator
160	11.705602	10.10.10.16	10.10.10.11	SMB	207	Session Setup AndX Response

Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on Interface 0
Ethernet II, Src: Microsoft_71:20:43 (08:15:5d:71:20:43), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Transmission Control Protocol, Src Port: 445, Dst Port: 44500, Seq: 134, Ack: 345, Len: 153

0000 ff ff ff ff ff ff 00 15 5d 71 20 43 00 00
0010 00 00 00 04 00 01 00 15 5d 71 20 43 0a 0a 0a
0020 00 00 00 00 00 00 0a 0a 0a 01
Address Resolution Protocol (Req), 28 bytes
Packets: 177 · Displayed: 177 (100.0%) · Profile: Default



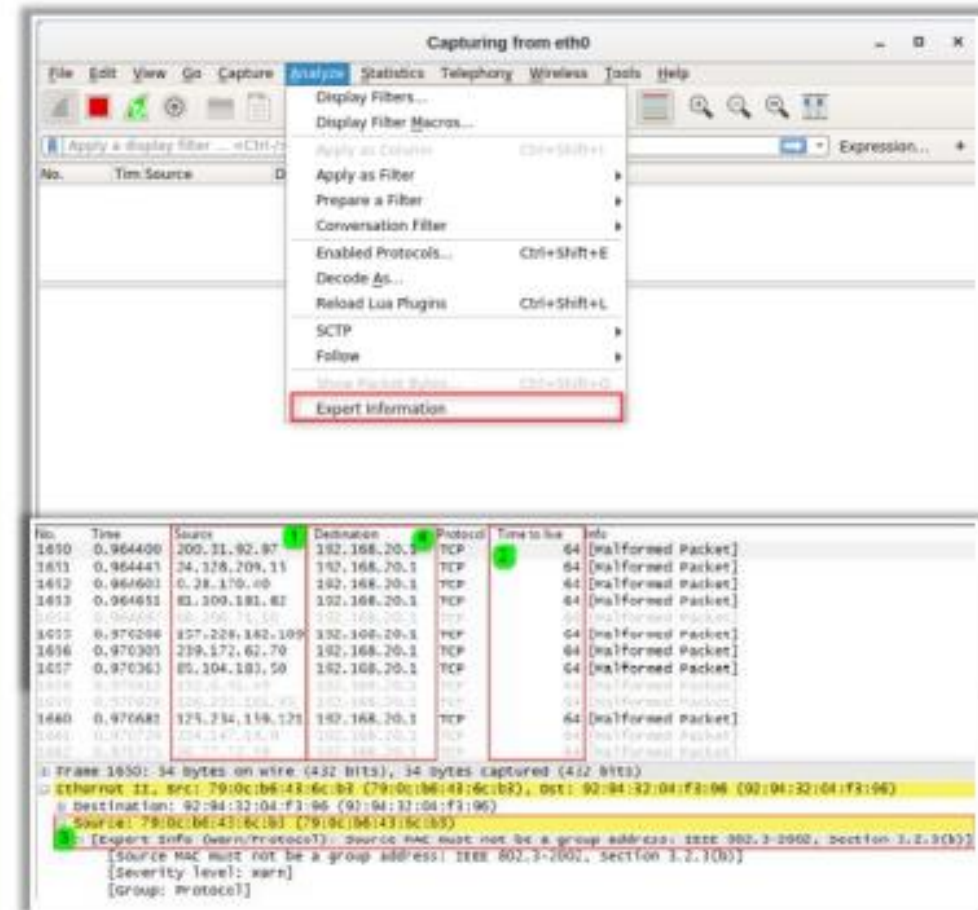
Analyze Traffic for Sniffing Attempts

- 1 Attackers sniff the network traffic for **sensitive** information
- 2 **Different techniques** may be used for sniffing the traffic depending on the type of network
- 3 Passive sniffing is used to sniff a **hub-based** network while active sniffing is used to sniff a **switch-based** network
- 4 An attacker uses **MAC flooding** and **ARP poisoning** to sniff network traffic, which can then serve as a basis for launching **man-in-the middle (MiTM) attacks**
- 5 **Identify sniffing attempts** by detecting the **signs** of MAC flooding and ARP poisoning using **Wireshark**



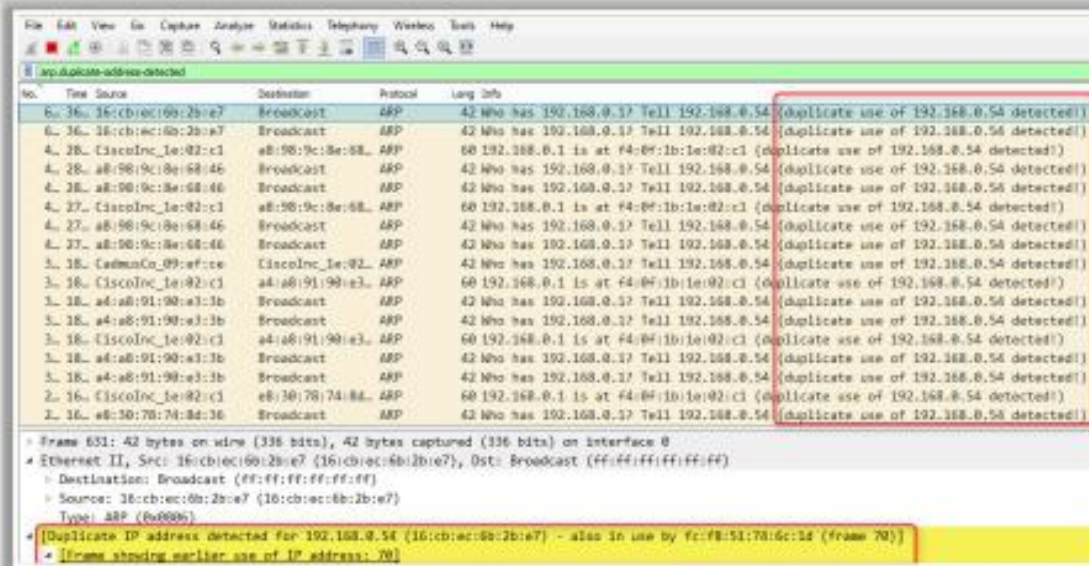
Analyze Traffic for MAC Flooding Attempt

- ❑ In a MAC flooding attack, the attacker connects to a port on the switch and **floods its interface** by sending a large volume of ethernet frames from various **fake MAC addresses**
- ❑ Wireshark considers MAC flooded packets as **malformed** packets
- ❑ Go to the **Analyze** menu on Wireshark and select **Expert Information** to view these malformed packets
- ❑ Analyze **source IP**, **destination IP**, and the **Time to Live (TTL) values** of the malformed packets while looking for **signs** of a MAC flooding attack on the network
- ❑ The screenshot below shows that the packets are originating from various IP **addresses (1)** and destined to the **same IP address 192.168.20.1 (4)** with the **same TTL values (2)** which indicates a MAC flooding attack



Analyze Traffic for ARP Poisoning Attempt

- ❑ In an ARP poisoning attack, the attacker's MAC address is associated with the IP address of the target host or a number of hosts in the target network
- ❑ Wireshark detects **duplicate IP addresses** on the ARP protocol with the warning message 'duplicate use of <IP address> detected'
- ❑ To locate duplicate IP address traffic use the filter: **arp.duplicate-address-detected**
- ❑ The screenshot shows here that the IP address 192.168.0.54 has two **different MAC addresses**, which has been detected by Wireshark as duplication



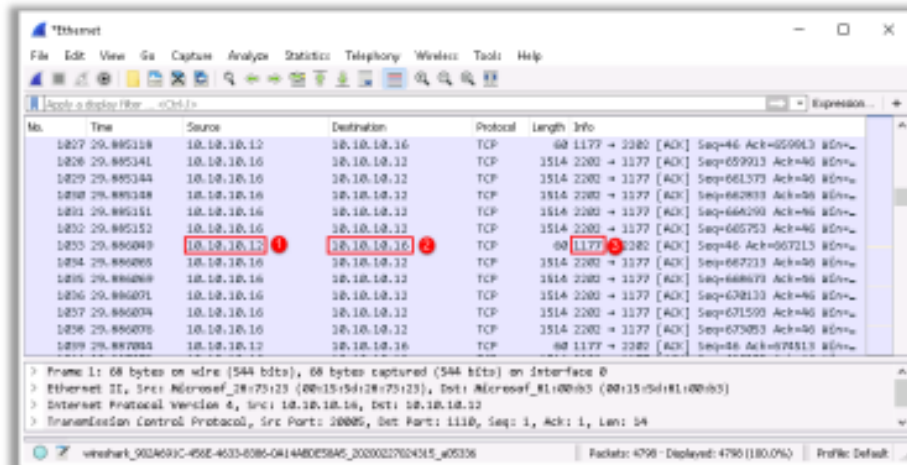
The screenshot shows a Wireshark packet capture with the filter 'arp.duplicate-address-detected'. The packet list table contains 16 entries, all of which are ARP broadcasts to the destination 192.168.0.54. The source MAC addresses are either 16:cb:ec:6b:2b:e7 or a4:a8:91:90:e3. The packet details pane for packet 16 shows the Ethernet II header and the ARP payload. The ARP payload section is highlighted in yellow and contains the following text:

```
> Frame 651: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
> Ethernet II, Src: 16:cb:ec:6b:2b:e7 (16:cb:ec:6b:2b:e7), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Destination: Broadcast (ff:ff:ff:ff:ff:ff)
> Source: 16:cb:ec:6b:2b:e7 (16:cb:ec:6b:2b:e7)
Type: ARP (0x0008)
[Duplicate IP address detected for 192.168.0.54 (16:cb:ec:6b:2b:e7) - also in use by fc:f8:51:7d:6c:1d (frame 70)]
[Frame showing earlier use of IP address: 70]
```

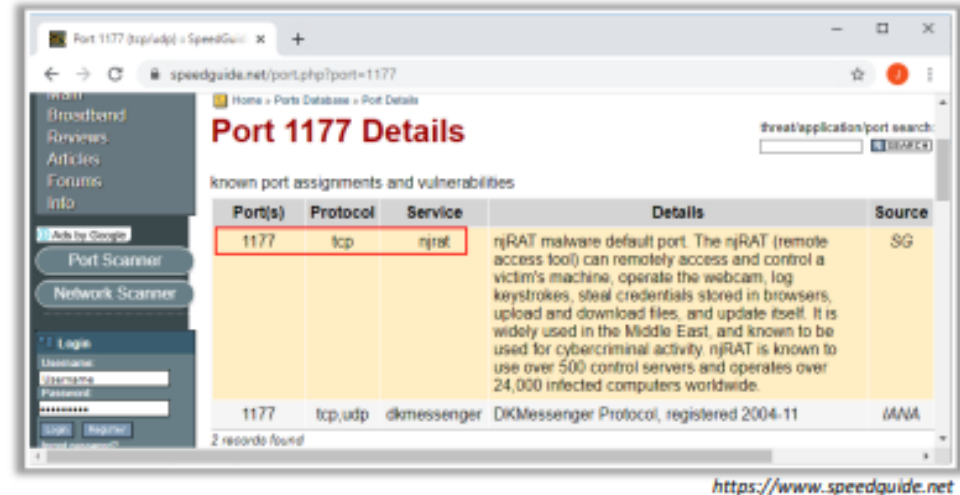


Analyze Traffic to Detect Malware Activity

- ❑ After a malware infects the target host, its activities on the network are often reflected in the ongoing traffic patterns which you can inspect and analyze via Wireshark
- ❑ In the screenshot below, the traffic analysis on Wireshark reveals that IP 10.10.10.12 (1) is trying to connect IP 10.10.10.16 (2) of the target host by using the port 1177 (3) which appears suspicious



- ❑ Search online databases to gather more information on any unusual/suspicious ports, IP addresses, or websites
- ❑ The port database of speedguide.net below shows the port 1177 to be malicious and used by the njRAT trojan



<https://www.speedguide.net>



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REFERENCES

- **Digital Forensics Essentials – 2021 by EC-Council**